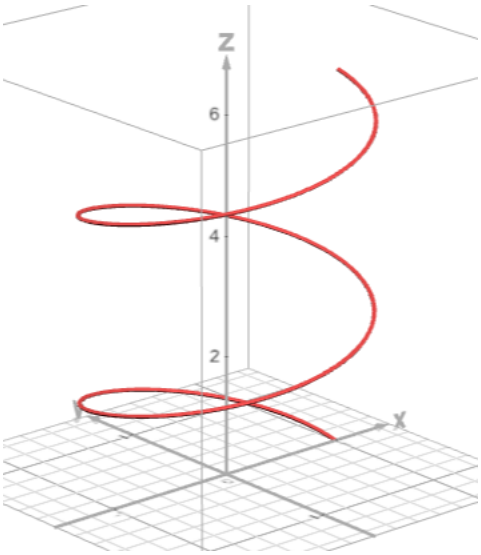
A. **Arclength.** If we add infinitesimal arclengths, we get totals arclength.

The **arclength** of a path $\vec{r}(t)$ with $a \leqslant t \leqslant b$ is

Example 1. Find the arclength of $\vec{r}(t) = \langle \cos(2t), \sin(2t), t \rangle$ with $0 \leqslant t \leqslant 2\pi$.



Solution. First we calculate the infinitesimal arclength $ds = \|\vec{r}'(t)\|\, dt$.

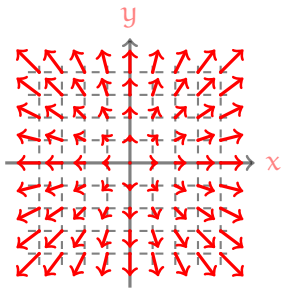
$ds =$

Next we integrate.

[arclength] $=$

B. **Vector Fields.** Imagine water flowing in the plane.

At each point in the plane of water, we indicate the velocity of water flow.



In our example, at point (x, y) the vector we root there is

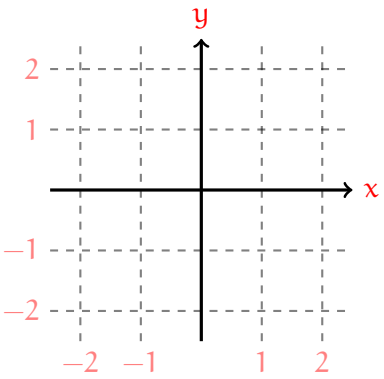
$\vec{F}(x, y) =$

This is an example of a **vector field**.

Example 2. Sketch the vector field at integer grid points in $[-2, 2] \times [-2, 2]$.

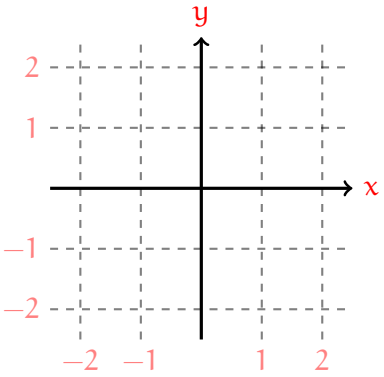
(a) $\vec{F}(x, y) = \langle 1/2, 1/2 \rangle$.

Solution. Rooted at each (x, y) we sketch the **constant** vector:



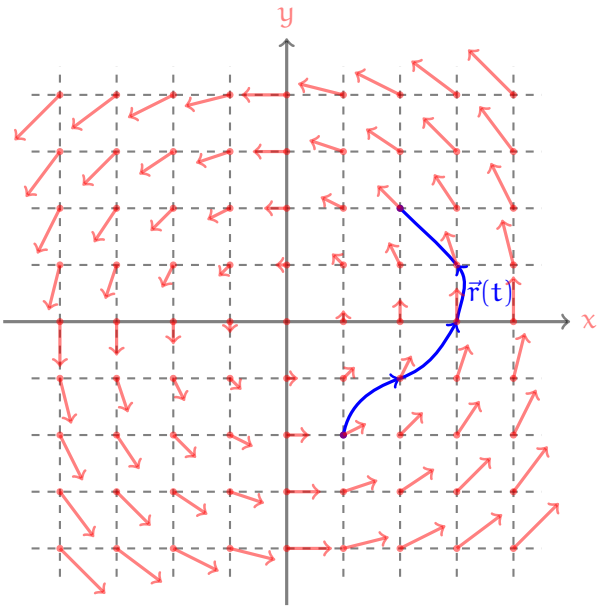
(b) $\vec{F}(x, y) = \langle -y/5, x/5 \rangle$.

Solution. Rooted at each (x, y) we sketch the vector:



C. **Vector Line Integrals.** We integrate vector fields over lines.

Consider this path $\vec{r}(t)$ in a fluid velocity vector field $\vec{F}$.

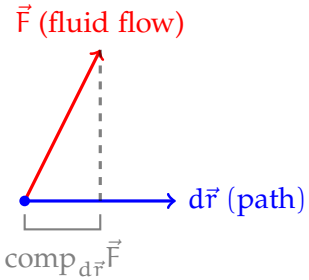


We will to measure the rate of fluid flowing along the curve.

An infinitesimal change in position along the curve is given by

$d\vec{r} =$

The extent to which $\vec{F}$ flows in the direction of $d\vec{r}$ is measured by

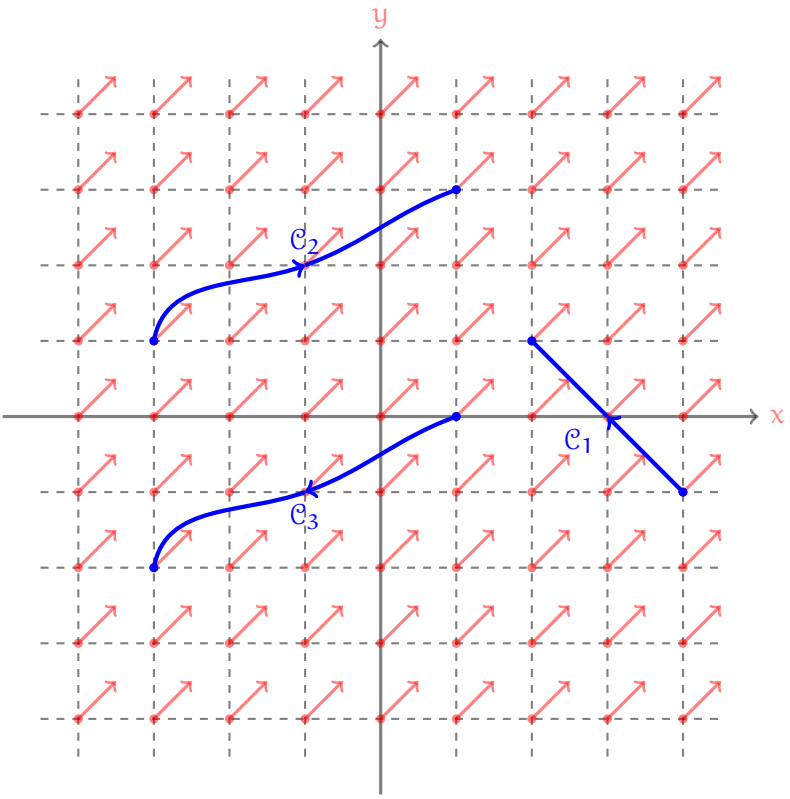


We sum this measurement over all points along the path.

The the **vector line integral** of $\vec{F}$ along path $\vec{r}(t)$ with $a \leqslant t \leqslant b$ is:

[rate of fluid flow along path] =

Example 3. Using the vector field $\vec{F}$ sketched below, decide whether the integral is positive, negative, or zero.



Solution. In each case we measure the extent to which fluid flows

$$\int_{\mathcal{C}_1} \vec{F} \cdot d\vec{r} =$$

$$\int_{\mathcal{C}_2} \vec{F} \cdot d\vec{r} =$$

$$\int_{\mathcal{C}_3} \vec{F} \cdot d\vec{r} =$$

Observe that specifiying a direction for our curve is essential.

A direction for a curve is called an:

If the direction is reversed, then the vector line integral is: